

REAL ESTATE ANALYTICS

Real Estate Price Analysis

Predicting Property Value from Location, Age & Amenities

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Python (EDA & Regression) · Tableau (Interactive Dashboards)

Objective

1

Understand Price Drivers

Identify which property and location factors most influence residential price per unit area.

2

Engineer & Transform Features

Extract time features from raw transaction dates and transform skewed variables to better fit a linear model.

3

Build a Predictive Model

Fit a multiple linear regression to estimate price from house age, transit distance, amenities, and location.

4

Communicate Findings Visually

Package the analysis into a reproducible notebook and an interactive Tableau dashboard for non-technical audiences.

Dataset Overview

Residential transactions — Sindian District, New Taipei City

414

Transactions

6

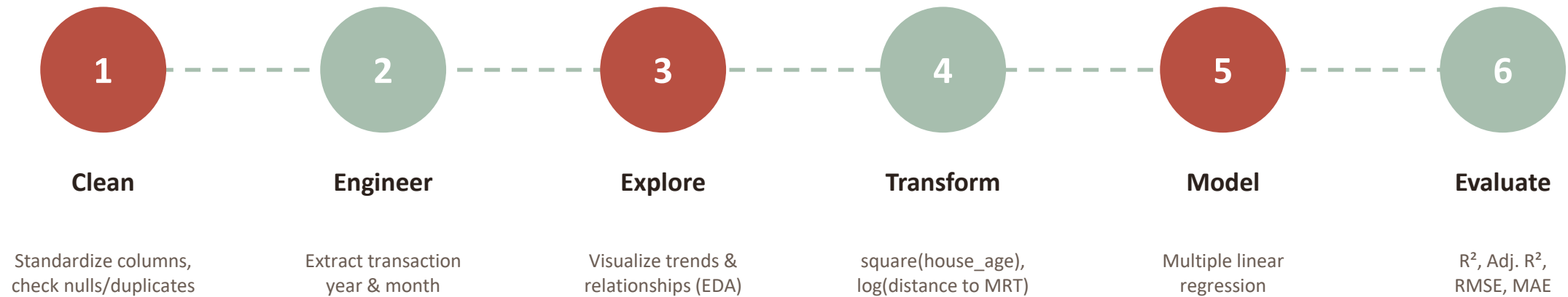
Predictive Features

0

Missing Values

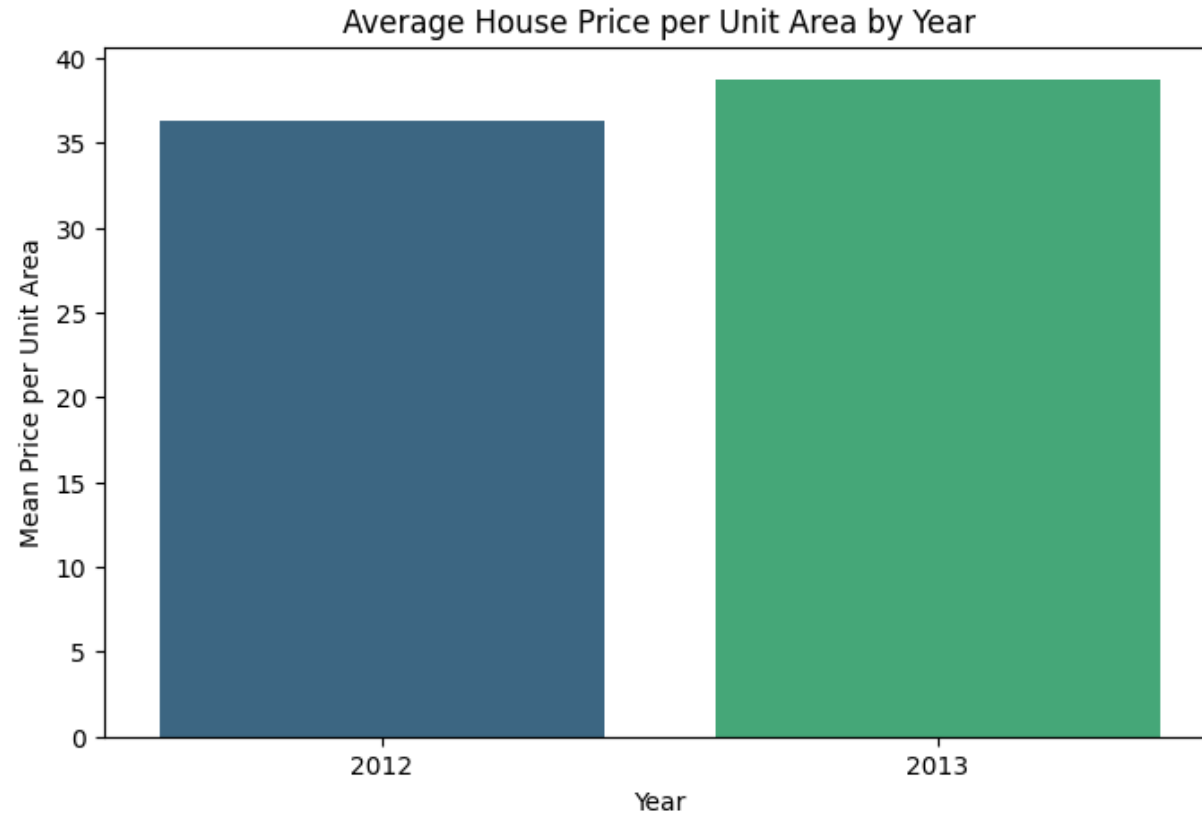
Feature	Description
transaction_date	Date of sale (decimal-year format)
house_age	Age of the property, in years
distance__mrt	Distance to nearest MRT (transit) station, meters
convenience_stores	Number of convenience stores within walking distance
latitude / longitude	Geographic coordinates
house_price_unit_area	Target — price per unit area

Methodology



Each stage builds on the previous one — every transformation and modeling choice is driven by what the EDA reveals.

EDA — Yearly Price Trend



Key Insight

Total transaction value per unit area in 2013 was roughly 2× that of 2012.

Average price per unit area also rose year-over-year, suggesting genuine market appreciation rather than just higher sales volume.

EDA — Amenity Effect



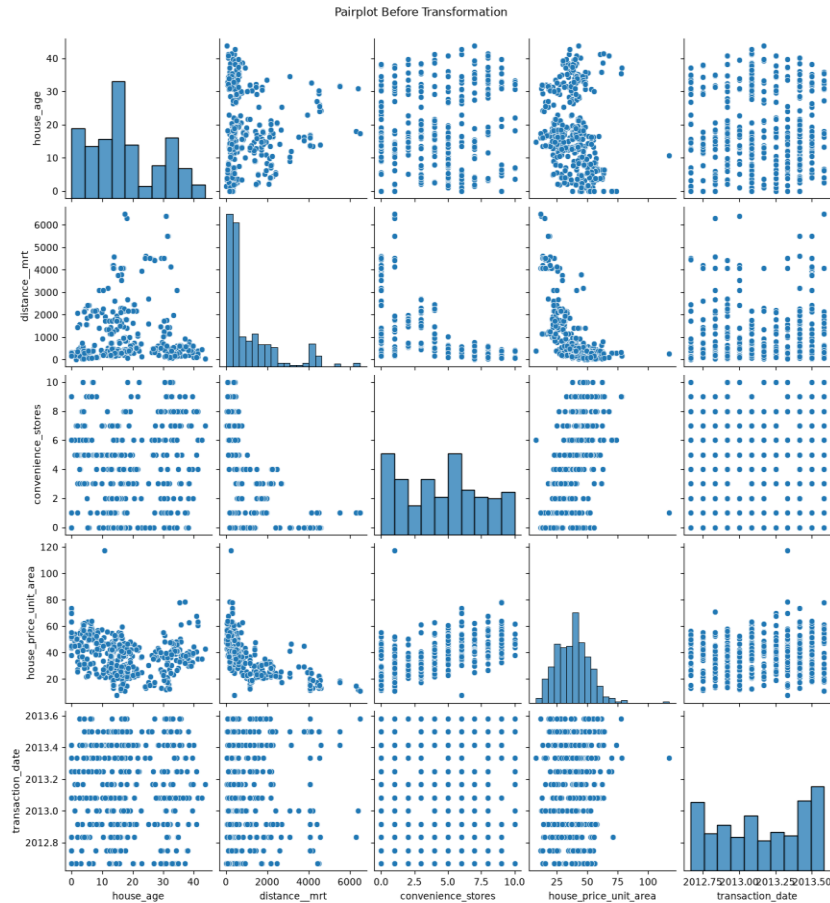
Key Insight

More nearby convenience stores tracks with higher price per unit area.

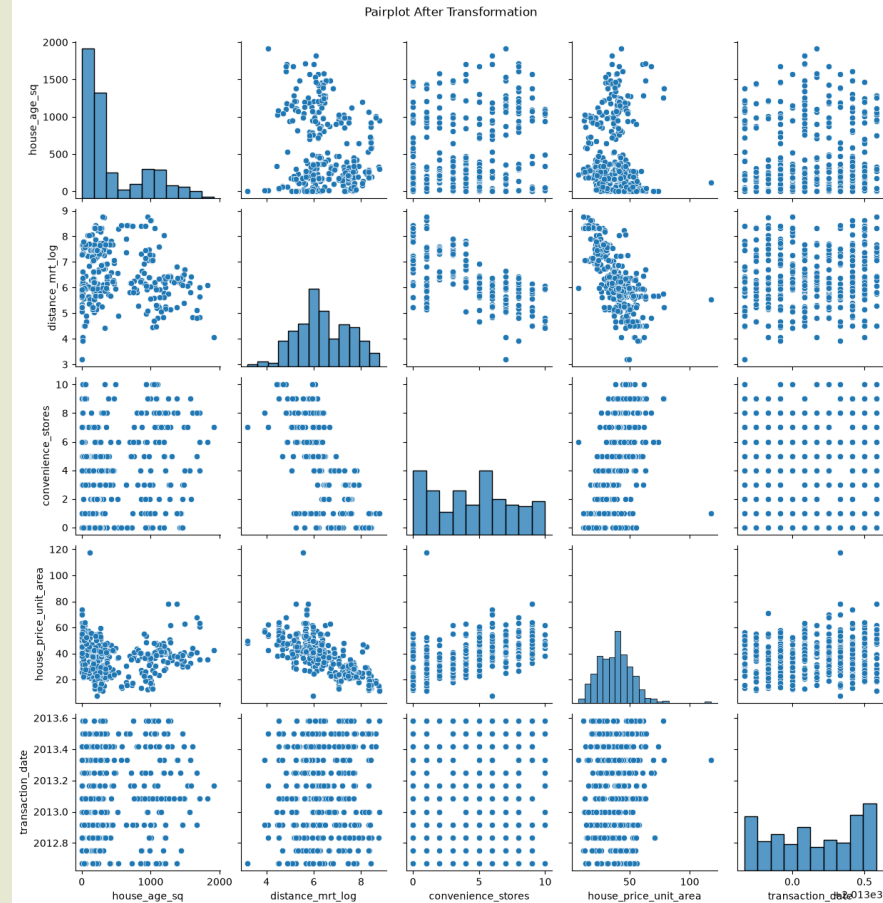
Walkability and amenity access are consistent value drivers across both 2012 and 2013 transactions.

Feature Transformation

Before



After



house_age_sq
= house_age^2

distance_mrt_log
= $\log(1 + \text{distance to MRT})$

Reduces skew and straightens relationships with price before fitting a linear model.

A Note on Location Features

The Limitation

Price doesn't vary linearly with latitude/longitude. Two coordinates close together can carry very different prices depending on what surrounds them — a pattern a straight-line model can't represent well.

The Decision

A tree-based model (Random Forest / Gradient Boosting) would capture this spatial signal better by splitting the region into local pockets.

For this linear regression, latitude and longitude were kept in rather than dropped — excluding them would discard real signal, and their coefficients remain directionally informative.

Model — Multiple Linear Regression

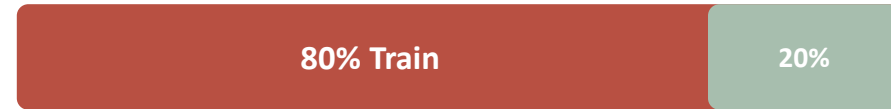
Predictor Features

- house_age_sq
- distance_mrt_log
- convenience_stores
- latitude
- longitude
- transaction_date

Target Variable

house_price_unit_area

Train / Test Split



Test set held out for unbiased evaluation. random_state=42 for reproducibility.

```
sklearn.linear_model.LinearRegression() + statsmodels.OLS() for coefficient significance (p-values)
```

Model Evaluation

Measured on the held-out 20% test set

0.716

R^2

Variance in price explained by the model

0.693

Adjusted R^2

R^2 penalized for number of predictors

6.91

RMSE

Typical prediction error, same units as price

4.45

MAE

Average absolute prediction error

Reading the Result

The model explains ~72% of the variation in price per unit area — solid for a 5-feature linear model on real transaction data. The remaining ~28% likely reflects the non-linear, spatial effects discussed on the previous slide, along with property-level factors (condition, floor, building quality) not captured in this dataset.

Coefficients & Slope Interpretation

Holding all other features constant · from statsmodels OLS on the full dataset

Feature	Coefficient	p-value	Interpretation
house_age_sqrt	-1.928	<0.001	Older properties (transformed) sell for less
distance_mrt_log	-5.938	<0.001	Farther from transit → notably lower price
convenience_stores	+0.490	0.011	Each nearby store adds modest value
latitude	+295.21	<0.001	Strong north-south price gradient
longitude	+24.71	0.480	<i>Not statistically significant in this model</i>

longitude's coefficient isn't statistically significant ($p = 0.48$) — a reminder that not every included feature earns its keep, and that spatial effects here are dominated by the north-south (latitude) axis rather than east-west.

Interactive Tableau Dashboard



EDA View

Explore price trends by year, month, and amenity count with filterable, interactive charts.

```
https://public.tableau.com/shared/CB3MH5W38?:display\_count=n&:origin=viz\_share\_link
```



ML Dashboard

Visualizes model predictions, coefficients, and prediction error against actual sale prices.

```
https://public.tableau.com/shared/5TW74TFJG?:display\_count=n&:origin=viz\_share\_link
```

Key Takeaways

- Transit distance and house age are the strongest, most significant price drivers in the model.
- Amenity access (convenience stores) adds modest but statistically significant value.
- Location coordinates are better suited to tree-based models — a clear next step for improving accuracy.
- The linear model explains ~71% of price variance ($R^2 = 0.708$) on unseen data — a solid, interpretable baseline.

Thank you

Muhammad Abdullah · Kaggle & GitHub portfolio